

Agents in AI

How software learned to act, not just answer

What Makes Software an Agent



Perceive

Read its environment.



Reason

Plan a course of action.



Act

Use tools to change things.

Assistant vs Agent

Assistant

Answers one prompt

Waits for you

Has no tools

Agent

Pursues a goal

Acts on its own

Calls tools

Core Capabilities



Planning

Break goals into steps.



Tool Use

Call APIs and code.



Memory

Recall past context.



Reflection

Critique and retry.

Why Now

2022

LLMs Arrive

tool use emerges

10x

Faster

early reports

1M+

Developers

building agents

From Rules to Agents

1980s

Rules

Hand-coded
logic.

2010s

ML

Learned patterns.

Today

Agents

Models that act.

Common Agent Patterns



ReAct

Reason,
then act.



Tool Use

Call
functions.



RAG

Retrieve
facts first.



Multi-Agent

Specialists
team up.

Multi-Agent Systems

Many narrow agents beat one generalist.

Each owns a clear role.

A coordinator routes the work.

They hand off context.

Promise vs Pitfalls

Promise

Automate real work

Scale expertise

Run all day

Pitfalls

Errors compound

Hard to predict

Needs guardrails

Takeaways

Software that acts, with human oversight